Final Exam

Name: ______________________________

Please circle your section number!

Section 1	Section 2	Section 3	Section 4
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- **There are 7 problems on the exam, some of them have multiple parts.**
- Read the problems carefully and budget your time wisely.
- If you are running out of time, please at least say *how* you would solve a given problem, preferably using notation, even if you do not have the time to do so. This may earn you partial credit.
- No calculators or other electronic devices, please. **Turn off your phone.**
- You do not have to carry out complicated numerical computations, but you should simplify your answer if it is possible with reasonable effort. In particular, geometric series must be evaluated.
- Please present your solutions in a clear manner. **Justify your steps.** (Unless it is explicitly stated that this is not required.) In problems 2.-7. numerical answers without explanation cannot get credit. Cross out the writing that you do not wish to be graded on.
- You can use the attached table on page 3 for the values of the $\Phi(x)$ function.
- On the second page you can find a list of named distributions with their PMF/PDF.
- You can use the empty pages for additional space for writing if necessary.

Problem	Points
1	/18
2	/12
3	/14
4	/14
5	/12
6	/20
7	/10
Total	/100

List of named distributions

$X \sim \text{Bernoulli}(p)$, PMF: $p_X(0) = 1 - p, \quad p_X(1) = p$

$X \sim \text{Binomial}(n, p)$, PMF: $p_X(k) = \binom{n}{k} p^k (1-p)^{n-k}, \quad 0 \le k \le n$

$X \sim \text{Geometric}(p)$, PMF: $p_X(k) = p(1-p)^{k-1}, \quad k = 1, 2, \ldots$

$X \sim \text{Poisson}(\lambda)$, PMF: $p_X(k) = \frac{\lambda^k}{k!} e^{-\lambda}, \quad k = 0, 1, 2, \ldots$

$X \sim \text{Hypergeometric}(N, N_A, n)$, PMF: $p_X(k) = \frac{\binom{N_A}{k}\binom{N-N_A}{n-k}}{\binom{N}{n}}, \quad 0 \le k \le n$

$X \sim \text{Uniform}(a, b)$, PDF: $f_X(x) = \begin{cases} \frac{1}{b-a} & \text{for } x \in [a, b], \\ 0 & \text{otherwise} \end{cases}$

$X \sim \text{Normal}(\mu, \sigma^2)$, PDF: $f_X(t) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(t-\mu)^2}{2\sigma^2}}$

$X \sim \text{Exponential}(\lambda)$, PDF: $f_X(t) = \begin{cases} \lambda e^{-\lambda x} & \text{for } x > 0, \\ 0 & \text{otherwise} \end{cases}$

$X \sim \text{Gamma}(r, \lambda)$, PDF: $f_X(t) = \begin{cases} \frac{1}{\Gamma(r)} \lambda^r x^{r-1} e^{-\lambda x} & \text{for } x > 0, \\ 0 & \text{otherwise} \end{cases}$

$(X_1, \ldots, X_k) \sim \text{Multinomial}(n, k, p_1, \ldots, p_k)$,
joint PMF: $p(a_1, \ldots, a_k) = \binom{n}{a_1, \ldots, a_k} p_1^{a_1} \cdots p_k^{a_k}$ for $a_1, \ldots, a_k \ge 0$ with $a_1 + \cdots + a_k = n$.

Table of values for $\Phi(x)$, the CDF of a standard normal random variable

	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990
3.1	0.9990	0.9991	0.9991	0.9991	0.9992	0.9992	0.9992	0.9992	0.9993	0.9993
3.2	0.9993	0.9993	0.9994	0.9994	0.9994	0.9994	0.9994	0.9995	0.9995	0.9995
3.3	0.9995	0.9995	0.9995	0.9996	0.9996	0.9996	0.9996	0.9996	0.9996	0.9997
3.4	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9998

1. **Short problems. (3 points each)** You do not need to justify your answers in this part of the exam.

   Suppose that X, Y are discrete random variables that take non-negative integer values with joint probability mass function $p(x, y)$. Express the following quantities in terms of $p(x, y)$.

   (Some answers might be in a form incorporating one or more sum. Make sure that the limits of the sums are properly identified. Your expressions should not include other functions besides $p(x, y)$.)

   (a) The probability mass function of X evaluated at 4.

   (b) $P(\max(X, Y) = 1)$

(Problem 1 continued)

(c) $P(X - Y = 1)$

(d) $E[X \cdot e^Y]$

(Problem 1 continued)

(e) The conditional probability mass function of X given $Y = 3$, evaluated at 6.

(f) $E[X|Y = 3]$

2. **(12 points)** We roll a fair die 20 times. Let X be the number of results we obtained exactly 3 times. (So if we get the following twenty numbers in some order: 1,1,1,1,2,2,2,3,3,3,3,3,4,4,4,5,6,6,6,6, then $X = 2$ because of the twos and the fours.) Find the expected value of X.
(You do not need to simplify your answer, but it should be a number, not a sum.)

3. Suppose that X is a non-negative random variable.

   (a) **(4 points)** We know that the mean of X is 100. What upper bound can you provide on $P(X \geq 150)$ using this (and only this) information?

   (b) **(4 points)** Suppose that we also learn (in addition to its mean) that the variance of X is 400. How can you improve on your upper bound on $P(X \geq 150)$?

(Problem 3 continued)

(c) **(6 points)** Suppose that we also learn (in addition to its mean and variance) that X is actually the sum of a 100 i.i.d. random variables. How would you estimate the probability $P(X \geq 150)$ now?

4. Suppose that we have the following information about the random variables X and Y:

$$E[X] = 1, \quad E[X^2] = 10, \quad E[Y] = 2, \quad E[Y^2] = 13, \quad \text{and} \quad \mathrm{Cov}(3X + 2Y, 2Y - 3) = -6.$$

(a) **(3 points)** Find the value of $E[X + 3Y]$.

(b) **(3 points)** Find $\mathrm{Var}(Y)$.

(Problem 5 continued)

(c) **(5 points)** Find the value of $\mathrm{Cov}(X, Y)$.

(d) **(3 points)** Find the correlation of X and Y.

5. **(12 points)** Let $X \sim \text{Exp}(1)$ and $Y \sim \text{Unif}(0,1)$ be independent. What is the probability density function of the random variable $Z = X + Y$?

   Make sure that you give the value of the PDF for all real numbers!

6. X is a continuous random variable with probability density function given by $f_X(x) = 12x^2(1-x)$ for $0 < x < 1$ and 0 otherwise. The conditional distribution of Y given $X = x$ (for $0 < x < 1$) is uniform on the interval $(x, 1)$.

   (a) **(4 points)** Find the conditional probability $P(Y > 0.8 | X = 0.6)$.

   (b) **(4 points)** Find the conditional expectation $E[Y|X]$.

(Problem 6 continued)

(c) **(4 points)** Find the probability density function of Y.

(d) **(4 points)** Find the conditional probability density function of X given $Y = y$. (Make sure to specify when the conditional PDF is defined.)

(Problem 6 continued)

(e) **(4 points)** Find two different single variable integrals for computing $E[Y]$ using the previous parts. (You do not need to evaluate the integrals.)

7. **(10 points)** You are given a fair die and a fair coin. You roll the die and flip the coin at the same time, and you repeat this once every minute. What is the probability that you roll your first 5 on the die exactly two minutes after seeing tails on the coin for the first time?

   (You do not need to simplify your answer, but it should be a number, not a sum.)

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